

# SAFETY DATA SHEET

Creation Date 16-June-2009

Revision Date 29-March-2024

Revision Number 3

## 1. Identification

**Product Name** Acetonitrile

**Cat No. :** 42879

**CAS-No** 75-05-8  
**Synonyms** AN; Methyl cyanide; Ethanenitrile

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

### Details of the supplier of the safety data sheet

**Company**  
**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**  
For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

## 2. Hazard(s) identification

### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                   |            |
|-----------------------------------|------------|
| Flammable liquids                 | Category 2 |
| Acute oral toxicity               | Category 4 |
| Acute dermal toxicity             | Category 4 |
| Acute Inhalation Toxicity         | Category 4 |
| Serious Eye Damage/Eye Irritation | Category 2 |

### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
Highly flammable liquid and vapor  
Harmful if swallowed, in contact with skin or if inhaled

Causes serious eye irritation



### Precautionary Statements

#### Prevention

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Keep container tightly closed

Ground/bond container and receiving equipment

Use explosion-proof electrical/ventilating/lighting/equipment

Do not breathe dust/fumes/gas/mist/vapours/spray

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Use only outdoors or in a well-ventilated area

Wear protective gloves/protective clothing/eye protection/face protection

Use only non-sparking tools

Take action to prevent static discharges

#### Response

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Call a POISON CENTER/ doctor if you feel unwell

Rinse mouth

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

Take off contaminated clothing and wash it before reuse

Wash contaminated clothing before reuse

#### Storage

Store in a well-ventilated place. Keep cool

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component    | CAS-No  | Weight % |
|--------------|---------|----------|
| Acetonitrile | 75-05-8 | <=100    |

## 4. First-aid measures

#### General Advice

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Inhalation

Remove to fresh air. If breathing is irregular or stopped, administer artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper

|                                        |                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | respiratory medical device. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                      |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Metabolism may release cyanide, which may result in headache, dizziness, weakness, collapse, unconsciousness, and possible death: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                               |

## 5. Fire-fighting measures

|                                         |                                                                                                                                  |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | Water may be ineffective, Do not use a solid water stream as it may scatter and spread fire                                      |
| <b>Flash Point</b>                      | 12.8 °C / 55 °F                                                                                                                  |
| <b>Method -</b>                         | No information available                                                                                                         |
| <b>Autoignition Temperature</b>         | 525 °C / 977 °F                                                                                                                  |
| <b>Explosion Limits</b>                 |                                                                                                                                  |
| <b>Upper</b>                            | 16 vol %                                                                                                                         |
| <b>Lower</b>                            | 3 vol %                                                                                                                          |
| <b>Oxidizing Properties</b>             | Not oxidising                                                                                                                    |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                         |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                         |

### Specific Hazards Arising from the Chemical

Flammable. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

Hydrogen cyanide (hydrocyanic acid). Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 2             | 3                   | 0                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Remove all sources of ignition. Take precautionary measures against static discharges. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Ensure adequate ventilation. Use personal protective equipment as required.                                                     |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                                                                                                                                                       |
| <b>Methods for Containment and Clean Up</b> | Remove all sources of ignition. Take precautionary measures against static discharges. Provide adequate ventilation. Use spark-proof tools and explosion-proof equipment. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Prevent product from entering drains. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. |
| <b>Storage.</b> | Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame. Flammables area. Incompatible Materials. Strong oxidizing agents. Strong acids. Reducing Agent. Bases.                                                                                                                                                                                                                                                                            |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component    | Alberta                                  | British Columbia    | Ontario TWAEV       | Quebec                                                                  | ACGIH TLV           | OSHA PEL                                                                                                                                                                                                          | NIOSH                                                                                                 |
|--------------|------------------------------------------|---------------------|---------------------|-------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Acetonitrile | TWA: 20 ppm<br>TWA: 34 mg/m <sup>3</sup> | TWA: 20 ppm<br>Skin | TWA: 20 ppm<br>Skin | TWA: 20 ppm<br>Ceiling: 10 ppm<br>Ceiling: 11 mg/m <sup>3</sup><br>Skin | TWA: 20 ppm<br>Skin | (Vacated) TWA: 40 ppm<br>(Vacated) TWA: 70 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 60 ppm<br>(Vacated) STEL: 105 mg/m <sup>3</sup><br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | IDLH: 137 ppm<br>IDLH: 25 mg/m <sup>3</sup><br>REL = 20 ppm (TWA)<br>REL = 34 mg/m <sup>3</sup> (TWA) |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments                                                                    |
|----------------|-------------------|-----------------|-----------------------------------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.35 mm         | As tested under EN374-3<br>Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European

Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371

#### Environmental exposure controls

No information available.

#### Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

## 9. Physical and chemical properties

|                                                |                                                           |                                          |
|------------------------------------------------|-----------------------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                                                    |                                          |
| <b>Appearance</b>                              | Colorless                                                 |                                          |
| <b>Odor</b>                                    | aromatic                                                  |                                          |
| <b>Odor Threshold</b>                          | 170 ppm                                                   |                                          |
| <b>pH</b>                                      | Not applicable                                            |                                          |
| <b>Melting Point/Range</b>                     | -46 °C / -50.8 °F                                         |                                          |
| <b>Softening Point</b>                         | No data available                                         |                                          |
| <b>Boiling Point/Range</b>                     | 81 - 82 °C / 177.8 - 179.6 °F                             | @ 760 mmHg                               |
| <b>Flash Point</b>                             | 12.8 °C / 55 °F                                           | <b>Method -</b> No information available |
| <b>Flammability (liquid)</b>                   | Highly flammable                                          | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                                            | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 3 vol %<br><b>Upper</b> 16 vol %             |                                          |
| <b>Autoignition Temperature</b>                | 525 °C / 977 °F                                           |                                          |
| <b>Decomposition Temperature</b>               | No data available                                         |                                          |
| <b>Water Solubility</b>                        | Miscible                                                  |                                          |
| <b>Solubility in other solvents</b>            | No information available                                  |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                           |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                            |                                          |
| Acetonitrile                                   | -0.34                                                     |                                          |
| <b>Vapor Pressure</b>                          | 97 mbar @ 20 °C                                           |                                          |
| <b>Density / Specific Gravity</b>              | 0.781                                                     |                                          |
| <b>Bulk Density</b>                            | Not applicable                                            | Liquid                                   |
| <b>Vapor Density</b>                           | 1.42                                                      | (Air = 1.0)                              |
| <b>Viscosity</b>                               | 0.36 cP at 20 °C                                          |                                          |
| <b>Particle characteristics</b>                | Not applicable (liquid)                                   |                                          |
| <b>Molecular Formula</b>                       | C <sub>2</sub> H <sub>3</sub> N                           |                                          |
| <b>Molecular Weight</b>                        | 41.05                                                     |                                          |
| <b>Explosive Properties</b>                    | Not explosive Vapors may form explosive mixtures with air |                                          |
| <b>Oxidizing Properties</b>                    | Not oxidising                                             |                                          |
| <b>Evaporation Rate</b>                        | 5.79 - (Butyl Acetate = 1.0)                              |                                          |

## 10. Stability and reactivity

|                            |                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>     | None known, based on information available                                                                        |
| <b>Stability</b>           | Stable under normal conditions.                                                                                   |
| <b>Conditions to Avoid</b> | Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.<br>Exposure to moisture. |

**Incompatible Materials** Strong oxidizing agents, Strong acids, Reducing Agent, Bases

**Hazardous Decomposition Products** Hydrogen cyanide (hydrocyanic acid), Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>)

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Information on expected route of exposure

**Inhalation** Avoid breathing vapors or mists. Harmful by inhalation.  
**Ingestion** May be harmful if swallowed.  
**Eyes** Avoid contact with eyes. Irritating to eyes. Vapor may cause irritation.  
**Skin** Avoid contact with skin. May cause irritation. Harmful in contact with skin. Prolonged skin contact may defat the skin and produce dermatitis.

### Acute Toxicity

#### Product Information Component Information

| Component    | LD50 Oral                                      | LD50 Dermal                                | LC50 Inhalation                                             |
|--------------|------------------------------------------------|--------------------------------------------|-------------------------------------------------------------|
| Acetonitrile | >= 450- <= 787 mg/kg (Rat), OECD Guideline 401 | >= 2000 mg/kg (Rabbit), OECD Guideline 402 | LC50 = 3587 ppm (6.022 mg/l) (Mouse) 4h, OECD Guideline 403 |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Irritating to eyes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component    | CAS-No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------|---------|------------|------------|------------|------------|------------|
| Acetonitrile | 75-05-8 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Metabolism may release cyanide, which may result in headache, dizziness, weakness, collapse, unconsciousness, and possible death: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

| Component    | Freshwater Algae | Freshwater Fish                                                                                                                                                                                                                     | Microtox                                                               | Water Flea |
|--------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------|
| Acetonitrile | Not listed       | LC50: = 1850 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 1000 mg/L, 96h static (Pimephales promelas)<br>LC50: 1600 - 1690 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 1650 mg/L, 96h static (Poecilia reticulata) | EC50 = 28000 mg/L 48 h<br>EC50 = 73 mg/L 24 h<br>EC50 = 7500 mg/L 15 h | Not listed |

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

| Component    | log Pow |
|--------------|---------|
| Acetonitrile | -0.34   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component              | RCRA - U Series Wastes | RCRA - P Series Wastes |
|------------------------|------------------------|------------------------|
| Acetonitrile - 75-05-8 | U003                   | -                      |

## 14. Transport information

### DOT

UN-No UN1648  
 Proper Shipping Name ACETONITRILE  
 Hazard Class 3  
 Packing Group II

### TDG

UN-No UN1648  
 Proper Shipping Name ACETONITRILE  
 Hazard Class 3  
 Packing Group II

### IATA

UN-No UN1648  
 Proper Shipping Name ACETONITRILE  
 Hazard Class 3  
 Packing Group II

### IMDG/IMO

UN-No UN1648  
 Proper Shipping Name ACETONITRILE  
 Hazard Class 3  
 Packing Group II

## 15. Regulatory information

### International Inventories

| Component | CAS-No | DSL | NDL | TSCA | TSCA Inventory | EINECS | ELINCS | NLP |
|-----------|--------|-----|-----|------|----------------|--------|--------|-----|
|-----------|--------|-----|-----|------|----------------|--------|--------|-----|

|              |         |   |   |   | notification -<br>Active-Inactive |           |   |   |
|--------------|---------|---|---|---|-----------------------------------|-----------|---|---|
| Acetonitrile | 75-05-8 | X | - | X | ACTIVE                            | 200-835-2 | - | - |

| Component    | CAS-No  | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------|---------|-------|----------|------|------|------|------|-------|-------|
| Acetonitrile | 75-05-8 | X     | KE-00067 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component    | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|--------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Acetonitrile | Part 1, Group A Substance Part 4 Substance           |                                                                            |                                           |

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Acetonitrile | -                                                                   | Use restricted. See entry 75. (see link for restriction details)              | -                                                                                                     |

**REACH links**
<https://echa.europa.eu/substances-restricted-under-reach>
**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component    | CAS-No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Acetonitrile | 75-05-8 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component    | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Acetonitrile | 75-05-8 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

Prepared By

Product Safety Department

Email: [chem.techinfo@thermofisher.com](mailto:chem.techinfo@thermofisher.com)



www.thermofisher.com

**Creation Date** 16-June-2009

**Revision Date** 29-March-2024

**Print Date** 29-March-2024

**Revision Summary** This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**